



NCI Protocol #: 10323

Patient ID:

Date:

1 of 13

Patient Name:
Date Collected:
Biopsy Site: **Abdomen**
Sample ID:
Telephone:
Tumor Content (%): **50_69%**

Patient DOB:
Referring Physician:
Primary Diagnosis: **Malignant melanoma**
MoCha ID:
Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Report Highlights

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Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	BRAF V600E Prognostic significance: None Diagnostic significance: None	atezolizumab + cobimetinib + vemurafenib ¹ binimetinib + encorafenib ¹ cetuximab + encorafenib ¹ cobimetinib + vemurafenib ¹ dabrafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ vemurafenib ¹ encorafenib	binimetinib + encorafenib ¹ cetuximab + encorafenib ¹ dabrafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ cobimetinib + vemurafenib encorafenib + panitumumab selumetinib vemurafenib	52
IIC	IDH1 R132C Prognostic significance: None Diagnostic significance: None	None	ivosidenib ¹	9

Public data sources included in relevant therapies: FDA¹, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants							
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
IDH1	p.(R132C)	c.394C>T	COSM28747	chr2:209113113	24.65%	NM_005896.3	missense
BRAF	p.(V600E)	c.1799T>A	COSM476	chr7:140453136	27.40%	NM_004333.4	missense

Comments:

N/A

Signed By: _____

Biomarker Descriptions

BRAF (B-Raf proto-oncogene, serine/threonine kinase)

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival^{1,2}. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF³. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead^{3,4,5}.

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)^{6,7,8,9,10}. Mutations at V600 belong to class 1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types^{4,11}. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions⁴. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and N581S/I⁴. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms^{12,13,14}. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in exon 11 as well as additional codons flanking V600 in the activation loop¹¹. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer^{7,10}. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types^{15,16,17,18,19}. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation^{5,15,17}.

Potential relevance: Vemurafenib²⁰ (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive⁴. BRAF kinase inhibitors including dabrafenib²¹ (2013) and encorafenib²² (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib²² is

Biomarker Descriptions (continued)

approved in combination with cetuximab²³ (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations⁴. Trametinib²⁴ (2013) and binimetinib²⁵ (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib²⁶ (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The PD-L1 antibody, atezolizumab²⁷, has also been approved in combination with cobimetinib and vemurafenib for BRAF V600 mutation-positive unresectable or metastatic melanoma. In 2018, binimetinib²⁸ was also granted breakthrough designation in combination with cetuximab and encorafenib for BRAF V600E mutant metastatic colorectal cancer. The pan-RAF kinase inhibitor, tovorafenib (DAY-101), was granted breakthrough therapy designation (2020) by the FDA for pediatric patients with advanced low-grade glioma harboring activating RAF alterations²⁹. The ERK inhibitor ulixertinib³⁰ was also granted a fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma³¹. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)^{32,33,34,35,36,37,38}. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported¹⁹.

IDH1 (isocitrate dehydrogenase (NADP(+)) 1)

Background: The IDH1 and IDH2 genes encode homologous isocitrate dehydrogenase enzymes that catalyze the conversion of isocitrate to α -ketoglutarate (α -KG)³⁹. The IDH1 gene encodes the NADP+ dependent cytoplasmic isocitrate dehydrogenase enzyme; IDH2 encodes the mitochondrial isoform.

Alterations and prevalence: Recurrent somatic mutations in IDH1 and IDH2 are mutually exclusive and observed in several malignancies including glioma, chondrosarcoma, intrahepatic cholangiocarcinoma, acute myeloid leukemia (AML), and myelodysplastic syndrome (MDS)⁴⁰. Recurrent IDH1 variants include predominately R132H/C plus other substitutions at lower frequencies. These gain of function variants confer neomorphic enzyme activity⁴¹. Although wild-type enzymatic activity is ablated, recurrent IDH1 variants catalyze the conversion of α -KG to D-2-hydroxyglutarate, an oncometabolite with diverse effects on cellular metabolism, epigenetic regulation, redox states, and DNA repair^{39,42}. Recurrent IDH1 mutations are present in 5-10% of patients with AML and 5% of patients with MDS^{43,44,45}. Recurrent IDH1 mutations are present in nearly 80% of lower grade diffuse gliomas^{7,10}.

Potential relevance: Ivosidenib⁴⁶ is FDA approved (2018) for the treatment of AML patients with IDH1 R132C/G/H/L/S variants⁴⁷. Ivosidenib has also been granted breakthrough designation (2020) for IDH1 mutated relapsed or refractory myelodysplastic syndrome (MDS)⁴⁸. IDH1 mutations are associated with inferior leukemia-free survival in primary myelofibrosis (PMF) and inferior overall survival in polycythemia vera (PV) but have been shown to confer improved prognosis in lower grade gliomas^{49,50,51}.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E

Relevant Therapy	FDA	NCCN	Clinical Trials*
dabrafenib	☒	☒	☒
dabrafenib + trametinib	☒	☒	☒
binimetinib + encorafenib	☒	☒	☒
cetuximab + encorafenib	☒	☐	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.03(006).

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
trametinib	☒	☒	☒
cobimetinib + vemurafenib	☒	☒	☒ (II)
vemurafenib	☒	☒	☒ (II)
atezolizumab + cobimetinib + vemurafenib	☒	☒	☒ (II)
encorafenib	☒	☒	☒ (I)
encorafenib + panitumumab	☒	☐	☒
selumetinib	☒	☐	☒
encorafenib, binimetinib, pembrolizumab	☒	☒	☒ (III)
sargramostim, nivolumab, ipilimumab	☒	☒	☒ (II/III)
abemaciclib + temuterkib	☒	☒	☒ (II)
atezolizumab, tiragolumab	☒	☒	☒ (II)
binimetinib, encorafenib	☒	☒	☒ (II)
dabrafenib, nivolumab, trametinib	☒	☒	☒ (II)
dabrafenib, trametinib	☒	☒	☒ (II)
encorafenib, binimetinib, nivolumab, ipilimumab	☒	☒	☒ (II)
HL-085, vemurafenib	☒	☒	☒ (II)
ipilimumab, binimetinib, nivolumab, encorafenib	☒	☒	☒ (II)
ipilimumab, nivolumab, encorafenib, binimetinib, dabrafenib, trametinib	☒	☒	☒ (II)
LXH254 , LTT-462, trametinib, ribociclib	☒	☒	☒ (II)
nivolumab, ipilimumab, radiation therapy	☒	☒	☒ (II)
RX208	☒	☒	☒ (II)
tocilizumab, ipilimumab, nivolumab	☒	☒	☒ (II)
vemurafenib, cobimetinib	☒	☒	☒ (II)
vemurafenib, cobimetinib, atezolizumab	☒	☒	☒ (II)
vemurafenib, cobimetinib, ipilimumab, nivolumab	☒	☒	☒ (II)
vemurafenib, dabrafenib	☒	☒	☒ (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
ASTX029	×	×	● (I/II)
bemcentinib, dabrafenib, pembrolizumab, trametinib	×	×	● (I/II)
dabrafenib, trametinib, antimalarial	×	×	● (I/II)
dabrafenib, trametinib, navitoclax	×	×	● (I/II)
encorafenib, binimetinib, palbociclib	×	×	● (I/II)
HH-2710	×	×	● (I/II)
mirdametinib, lifirafenib	×	×	● (I/II)
tazemetostat, trametinib, dabrafenib	×	×	● (I/II)
vemurafenib, metformin hydrochloride	×	×	● (I/II)
ABM-1310	×	×	● (I)
AZD-0364	×	×	● (I)
BGB-3245	×	×	● (I)
binimetinib, nivolumab, encorafenib	×	×	● (I)
CART-GD2, vemurafenib	×	×	● (I)
cobimetinib, belvarafenib	×	×	● (I)
cobimetinib, vemurafenib	×	×	● (I)
DCC-3116, trametinib	×	×	● (I)
E6201, dabrafenib	×	×	● (I)
JSI-1187, dabrafenib	×	×	● (I)
KIN-2787	×	×	● (I)
nilotinib, dabrafenib, trametinib	×	×	● (I)
PF 07284890, binimetinib	×	×	● (I)
RMC-4630	×	×	● (I)
RO-7276389, cobimetinib	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

IDH1 R132C

Relevant Therapy	FDA	NCCN	Clinical Trials*
ivosidenib	☐	☐	☒ (II)
ivosidenib, nivolumab	☒	☒	☒ (II)
olaparib	☒	☒	☒ (II)
olaparib, ceralasertib	☒	☒	☒ (II)
olaparib, durvalumab	☒	☒	☒ (II)
talazoparib	☒	☒	☒ (II)
HMPL-306	☒	☒	☒ (I)
KY100001	☒	☒	☒ (I)
LY-3410738	☒	☒	☒ (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

BRAF V600E

NCT ID	Title	Phase
NCT04657991	A Phase III, Randomized, Double-blind Study of Encorafenib and Binimetinib Plus Pembrolizumab Versus Placebo Plus Pembrolizumab in Participants With BRAF V600E/K Mutation-Positive Metastatic OR Unresectable Locally Advanced Melanoma	III
NCT03235245	Combination of Targeted Therapy (Encorafenib and Binimetinib) Followed by Combination of Immunotherapy (Ipilimumab and Nivolumab) vs Immediate Combination of Immunotherapy in Patients With Unresectable or Metastatic Melanoma With BRAF V600 Mutation : an EORTC Randomized Phase II Study (EBIN)	II
No NCT ID	A Phase IA/B Open-label Study to Evaluate Safety, Pharmacokinetics, and Preliminary Clinical Activity of R07276389 alone and in Combination with Cobimetinib in Participants with BRAF-V600 Mutation-positive Advanced Solid Tumor or BRAF-V600 Mutation-positive Melanoma with Central Nervous System Metastases	I
NCT04511013	A Randomized Phase II Trial of Encorafenib + Binimetinib + Nivolumab vs Ipilimumab + Nivolumab in BRAF-V600 Mutant Melanoma With Brain Metastases	II
NCT04417621	A Randomized, Open-label, Multi-arm, Two-part, Phase II Study to Assess Efficacy and Safety of Multiple LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic BRAFV600 or NRAS Mutant Melanoma	II

Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT03754179	A lead-in Phase I Followed by a Phase II Clinical Trial on the Combination of Dabrafenib, Trametinib and the Autophagy Inhibitor Hydroxychloroquine in BRAF/MEK Inhibitor-pretreated Patients With Advanced BRAF V600 Mutant Melanoma	I/II
NCT02872259	A Phase Ib/II Randomised Open Label Study of BGB324 in Combination With Pembrolizumab or Dabrafenib/Trametinib Compared to Pembrolizumab or Dabrafenib/Trametinib Alone, in Patients With Advanced Non-resectable (Stage IIIC) or Metastatic (Stage IV) Melanoma.	I/II
NCT03340129	A Phase II, Open Label, Randomised, Controlled Trial of Ipilimumab and Nivolumab With Concurrent Intracranial Stereotactic Radiotherapy Versus Ipilimumab and Nivolumab Alone in Patients With Melanoma Brain Metastases.	II
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT03898908	Phase II, Multicentre Clinical Trial to Evaluate the Activity of Encorafenib and Binimetinib Administered Before Local Treatment in Patients With BRAF Mutant Melanoma Metastatic to the Brain.	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT02231775	Neoadjuvant and Adjuvant Dabrafenib and Trametinib in Patients With Clinical Stage III Melanoma (Combi-Neo)	II
No NCT ID	A Single-Arm, Multi-Center Phase II Clinical Study To Evaluate The Effectiveness And Safety Of HL-085 Capsules Combined With Vimurafenib Tablets In The Treatment Of Patients With Advanced Melanoma With BRAF V600E/V600K Mutations	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT02968303	Phase II Study With COmbination of Vemurafenib With Cobimetinib in B-RAF V600E/K Mutated Melanoma Patients to Normalize LDH and Optimize Nivolumab and Ipilimumab therapy	II
NCT01989585	Phase I/II Study of Dabrafenib, Trametinib, and Navitoclax in BRAF Mutant Melanoma (Phase I and II) and Other Solid Tumors (Phase I Only)	I/II
NCT04655157	A Multi-Center Phase I/II Open Label Study to Evaluate Safety and Efficacy in Participants With Metastatic BRAF-mutant Melanoma Treated With Encorafenib With and Without Binimetinib in Combination With Nivolumab and Low-dose Ipilimumab. (QUAD 01: Quadruple Therapy in Melanoma)	I/II
NCT04557956	Phase I/II Study of an EZH2 Inhibitor (Tazemetostat) in Combination With Dual BRAF/MEK Inhibition in Patients With BRAF- Mutated Metastatic Melanoma Who Progressed on Prior BRAF/MEK Inhibitor Therapy	I/II
NCT01638676	A Phase I/II Trial of Vemurafenib and Metformin to Unresectable Stage IIIC and Stage IV BRAF.V600E+ Melanoma Patients	I/II
NCT04190628	A Phase I, First-In-Human, Multicenter, Open-Label Study of ABM-1310, Administered Orally in Adult Patients With Advanced Solid Tumors	I
No NCT ID	Phase I Study of Safety and Immune Effects of an Escalating Dose of Autologous GD2 Chimeric Antigen Receptor-Expressing Peripheral Blood T cells in Patients with Metastatic Melanoma	I
NCT05003622	Multicenter, Open-label, Phase I Study Investigating the Safety and Tolerability of Encorafenib Monotherapy in BRAF V600E-mutated Chinese Patients With Advanced Metastatic Solid Tumors	I

Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
No NCT ID	Phase I study of the safety, tolerability, pharmacokinetics and preliminary efficacy of RX208 in patients with advanced malignant solid tumors	I
NCT04722575	A Phase II Randomized Non-Comparative Study, With Neoadjuvant Plus Adjuvant Therapy With Combination Or Sequence Of Vemurafenib, Cobimetinib, And Atezolizumab In Patients With High-Risk, Surgically Resectable BRAF Mutated And Wild-Type Melanoma	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT05114603	An Open-label, Multicenter Phase II Clinical Study to Evaluate Safety, Efficacy and PK of HLX208 for Advanced Melanoma With BRAF V600 Mutation	II
NCT03220035	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of Vemurafenib in Patients With Tumors Harboring Braf V600 Mutations	II
NCT04720768	A Phase IB Study to Evaluate the Safety, Tolerability and Preliminary Efficacy of the Combination of Encorafenib, Binimetinib and Palbociclib in Patients With BRAF-mutant Metastatic Melanoma (The CELEBRATE Study)	I/II
NCT04249843	A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumor	I
NCT04741997	A Randomized Pilot Trial of Adjuvant Therapy Based on Pathologic Response After Neoadjuvant Encorafenib and Binimetinib in Advanced Melanoma	I
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT03543969	Pilot Study of Adaptive BRAF-MEK Inhibitor Therapy for Advanced BRAF Mutant Melanoma	I
NCT03332589	A Phase I Study of E6201 Plus Dabrafenib for the Treatment of Central Nervous System Metastases (CNS) From BRAF V600-Mutated Metastatic Melanoma	I
NCT03781219	A Phase I, Single Arm, Dose Escalation Study to Evaluate Safety, Pharmacokinetics and Preliminary Efficacy of HL-085 Plus Vemurafenib in Patients With BRAF V600 Mutant Advanced Solid Tumor	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I
NCT04903119	A Phase I Study of Nilotinib in Combination With Dabrafenib and Trametinib in BRAF V600 Mutant Metastatic Melanoma After Progression on BRAF/MEK Inhibition	I
NCT04543188	A Two-Part, Phase Ia/b, Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF 07284890 (ARRY 461) in Participants with Braf V600 Mutant Solid Tumors with and without Brain Involvement	I
NCT03554083	Vemurafenib, Cobimetinib, Atezolizumab, and Tiragolumab in Treating Patients With High-Risk Stage III Melanoma	II
NCT04940299	A Phase II Study to Assess the Safety and Efficacy of Tocilizumab in Combination With Ipilimumab and Nivolumab in Patients With Advanced Melanoma, Non-Small Cell Lung Cancer, or Urothelial Carcinoma	II

Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT04534283	A Phase II Basket Trial of an ERK1/2 Inhibitor (LY3214996) in Combination With Abemaciclib for Patients Whose Tumors Harbor Pathogenic Alterations in BRAF, RAF1, MEK1/2, ERK1/2, and NF1	II
NCT02910700	A Phase II Study of the TRiplet Combination of Dabrafenib, Nivolumab, and Trametinib in Patients With Metastatic Melanoma (TRIDeNT)	II
NCT03563729	Efficacy Of Immunotherapy In Melanoma Patients With Brain Metastases Treated With Steroids	II
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT03239015	Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event	II
NCT04892017	A Phase I, First-in-Human Study of DCC-3116 as a Single Agent and in Combination With Trametinib in Patients With Advanced or Metastatic Solid Tumors With RAS or RAF Mutations.	I
NCT02339571	Randomized Phase II/III Study of Nivolumab Plus Ipilimumab Plus Sargramostim Versus Nivolumab Plus Ipilimumab in Patients With Unresectable Stage III or Stage IV Melanoma	II/III
NCT04913285	A Phase I/Ib Open-Label, Multicenter, Two Part Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Antitumor Activity of KIN-2787 in Subjects With BRAF Mutation Positive Solid Tumors	I
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT04198818	A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors	I/II
NCT03634982	A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors	I
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I

IDH1 R132C

NCT ID	Title	Phase
NCT04056910	Phase II Study of IDH1 Inhibitor Ivosidenib and Nivolumab In IDH1 Mutant Gliomas And Advanced Solid Tumors	II
NCT03212274	A Phase II Study of the PARP Inhibitor Olaparib (AZD2281) in IDH1 and IDH2 Mutant Advanced Solid Tumors	II
NCT04195555	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) - Phase II Subprotocol of AG-120 (Ivosidenib) in Patients With Tumors Harboring IDH1 Mutations	II
NCT04521686	Study of LY3410738 Administered to Patients With Advanced Solid Tumors With IDH1 Mutations.	I

Clinical Trials Summary (continued)

IDH1 R132C (continued)

NCT ID	Title	Phase
NCT03878095	A Phase II Study of Olaparib and AZD6738 in Isocitrate Dehydrogenase (IDH) Mutant Solid Tumors	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT04906473	A Phase I Clinical Study of Single-arm, Open, Single/Multiple Dose Escalation and Dose Extension to Assess the Safety, Tolerability, Pharmacokinetic, Pharmacodynamics and Efficacy of KY100001 in Patients With Advanced Solid Tumors	I
NCT03991832	A Phase II Study of Olaparib and Durvalumab (MEDI 4736) in Patients With IDH-Mutated Solid Tumors	II
NCT04762602	A Multicenter, Open-Label, Phase I Study Evaluating the Safety and Tolerability of HMPL-306 in Subjects With Advanced or Metastatic Solid Tumors With IDH Mutations	I

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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